Prasad S. Gawas

Electronics And Communication Engineer

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LinkedIn: https://www.linkedin.com/in/prasad-gawas-a2a3631a0 ◆ **WWW:** Bold Profile

Professional Summary

Electronics Engineer skilled in **PCB design, component-level debugging, and hardware testing**, with strong programming proficiency in Python and C++. Familiar with PCB Debugging, and PCBA test workflows. Experienced with **PLC logic development** and **microcontroller architecture & firmware programming**. Motivated to contribute to **PCBA Testing, Test Automation, and Hardware Validation roles**.

Technical Competencies

Hardware & PCB

- PCB Design (KiCad), Circuit Schematic Interpretation
- Component-Level Debugging, Signal Tracing, Hardware Bring-up
- Oscilloscope, Multimeter, Logic Analyzer Usage
- Understanding of SMT, THT, Reflow, Wave Soldering, AOI & ICT workflows

PCBA Testing & Validation

- Familiar with **In-Circuit Testing (ICT)** concept
- Understanding of **Flying Probe** test methodology
- Knowledge of **Boundary Scan (JTAG)** & TAP chain configuration
- Test Coverage Planning & Net-Level Fault Isolation

Programming & Automation

- **Python**, **C/C++**, **VB.NET**, **LabView, LabWindows/CVI**, Embedded C
- Test Sequencing, Data Logging, Script-Based Automation
- Debugging Hardware-Software Interaction

Microcontrollers & Processors

- Architecture & programming of **8051, ARM Cortex, PIC** families
- Assembly-Level Programming, Peripheral Interfacing & Firmware Debugging

Work History

Apprenticeship, 09/2025 - Current
Technology Dept, Siemens – Verna, India

- Developed software to evaluate cycle time and track downtime for DEK SMT stencil printer.
- Created Bulk Program Generators for Ersa Three/20 and Seho Reflow machines with precise values mapping for heating, cooling, convection, and CBS/Park position parameters.
- Implemented Python-based tool to analyze and report daily operational downtime of production lines.

- Refined Live Parameters Comparison tool to ensure accuracy in seho reflow and Wave soldering machines.
- Developing an Alert Generation system which indicates and alerts authority about missing program / corrupted program and machine errors as per the daily production schedule.
- Implemented ETHON.AI to enhance PCBA inspection efficiency on THT lines through model training.

Internship, 07/2024 - 08/2024
Siemens – Verna, India

- Designed and implemented software for the Saki AOI machine to evaluate pass/fail results and diagnose defects in SMT-manufactured PCBs, achieving a 30% increase in operational efficiency without disturbing the existing system and server database.
- Along with 3 engineering interns to develop scalable automation solutions for machine processes, stencil inventory, and shelf management, ensuring adherence to business objectives and regulatory standards.
- Developed custom on-demand software for the Conformal Coating Machine to verify PCB-palette configurations, automate coating processes, and conduct final image-based validation, with on-site report generation for efficient process tracking.
- Acquired in-depth knowledge of robotic arms, SMT-THT processes, and machines, along with hands-on experience in cobot programming.

Skills

- Machine Learning
- Robotics and mechatronics
- Additive Manufacturing
- Game Development

Education

Bachelor in Engineering: Electronics & Communication Engineering
Agnel Institute of Technology And Design - Assagao Goa

- Honours in IoT

HSSC: Science
Shri Bhumika Higher Secondary School - Parye, Goa

SSC: Goa Board
Shri Ganesh Vidhyalaya - Harvalem, Goa

Languages

English (Proficient)
Hindi (Proficient)
Konkani (Native)
Marathi (Proficient)

Hobby

Stargazing and exploring constellations with telescope & sky-tracking apps